

Experiencing Presence in a Virtual Reality Music Video

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ABSTRACT

As some music artists begin to explore virtual reality (VR) as a medium for their music videos, it was found challenging to direct people's attention using traditional film techniques. These must, therefore, be innovative or alternative methods established. To evaluate the effectiveness of attention cues and their correlation of eliciting a sense of presence, a music video was created in a virtual environment (VE). Two design iterations were completed. The first investigated the size, colour, and movement pattern of the visual cues and the second examined if the cue was perceived as a part of the virtual world. It was found that the cue should have a vibrant yet not necessarily contrasting colour, depending on the background (here pink on purple), and have animate motion, giving a more coherent perception of the world. This study investigated the user's level of presence experiencing the music video in VR, compared to a smartphone. The results demonstrate that the feeling of presence in the VR is significantly higher.

Keywords: Virtual Reality, Music Videos, Immersion, Presence, Attention Cues

1 INTRODUCTION

In traditional films, cinematography is utilised to guide the attention of the audience [1], aiding to perceive the narrative as the film-makers intend. Conveying a story is not only fundamental for films but also music videos, since these function as music artists' personal messages [5]. A lack of guidance can lead to confusion or misunderstanding of the explicit/implicit/symptomatic meaning [1].

Music videos are dominating on the streaming service YouTube, which reaches more people (18-49 years) than any broadcast or cable TV network alone on mobile phones [12]. Artists within the music industry thus have a wide audience to entertain and a fan-base to maintain. Being amongst the many, an emotional connection to the artist can at times be drowned in noise. Hence, investigating a way to maintain or (re)build the connection between the artist and audience is entailed.

To captivate their audience, music artists use MVs as a medium, to not only aurally but also visually personalise their appeal [5]. Producing MVs, have therefore become a greater part of the artist's creative process. This gives rise to speculation that including the audience in the MV, allowing them to immerse themselves in the artist's world. Such immersive experience has proven to be achievable with VR. Working with VR creates new challenges for MV producers. Cinematography, editing, and mise-en-scène are traditional film techniques used to guide the attention of the audience to specific areas on the screen, aiding them to follow the narrative [1]. These methods are, however, not directly applicable to VR productions. Modifications of existing methods or finding others is, therefore, necessary to obtain the similar effect.

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Speculations on the immersive abilities of an MV made for VR, the following research question was devised: *How can an MV within a VE utilising visual trends and attention cues create presence?*

2 RELATED WORK

It is not a novel phenomenon, creating MVs for VR. Taryn Southern with "Life Support", MUSE with "Revolt", and Night Club with "Show It 2 Me", are a few examples of MVs that can be seen in VR or as a 360° video. Hence, there are already artists who explore VR as a medium to create immersive audiovisual art.

In the studies of M. Slater et al. [14] and N. Nilsson et al. [11], 'immersion' is defined as the feeling of being surrounded by something or in the case of VR, having a feeling of *being there*. Nilsson et al. furthermore present three sub-categories of immersion; 1) system immersion, 2) narrative immersion, and 3) challenge-based immersion. With the focus on making an immersive MV, system- and narrative immersion will be in focus. System immersion depends on the system's ability to preserve the fidelity of equivalent real-world sensory modalities. The better these are preserved by displays and tracking, the more immersive of a system [11]. Narrative immersion covers how the user gets involved in the narrative and accepts the diegesis as real [11]. This can be achieved by e.g. incorporating a story-line or a theme for the world, which is up to the artist to define.

While immersion is centred around imitating physical stimulations, the psychological aspect of feeling as if belonging to the artist's world is referred to as 'presence' [13]. Presence can be achieved by implementing high-quality visuals and audio, as well as using an ergonomic head-mounted display (HMD), allowing for a 360° experience [17, 7]. A heavy and/or ill-fitted HMD can cause general discomfort and restrict the user's movement, removing focus from the experience. Additionally, while a wide field-of-view (FoV) can increase the sense of presence, it simultaneously increases the likelihood of cybersickness [2]. Cybersickness is related to exposure to VR, where symptoms such as dizziness, nausea, and headaches are generated, and is a decreasing factor for presence, thus should be avoided [2].

With the goal of immersing the viewer in the narrative of the MV, guiding their attention towards a specific region-of-interest (ROI), can be done by implementing visual cues. Speicher et al. [15] evaluated seven different cues, categorised by explicit/implicit, diegetic/non-diegetic and if they allow or limit interaction, as seen in Figure 1. The cues were forced rotation (FR), object to follow (OF), person to follow (PF), manipulation of objects (OM), environmental manipulation (EM), small gestures (SG), and big gestures (BG). They concluded that the most successful cue to guide the attention while not breaking the sense of presence, is OF. It is worth mentioning that OF is presented as non-diegetic in Figure 1, it can be integrated as a part of the diegesis, giving an undisturbed and compelling impression of the VE [10, 15].

The movement of the OF can be explained by the zone-division system (Figure 2), that is comparable to traditional framing [8]. The "frames" A, B, C, and D refer to the areas in front, left, right, and behind the viewer, respectively. Depending on the placement of

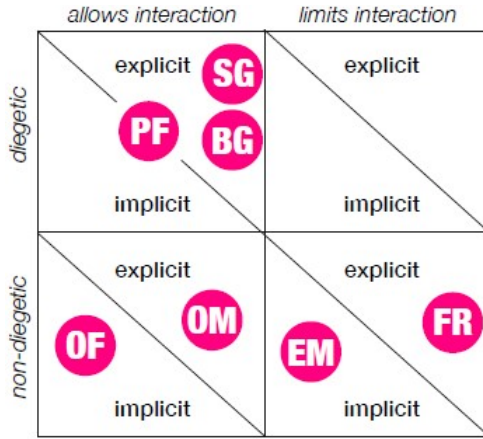
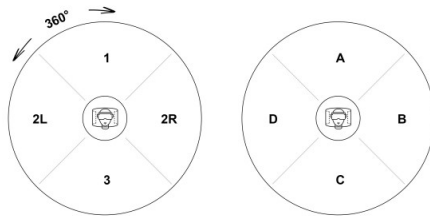


Figure 1: The categorisation of the visual cues evaluated by Speicher et al. [15].

the ROI, the division zones 1, 2.L, 2.R, 3, rotate around the user, where the ROI is always in zone 1 [8]. Within ROIs, there are usually placed objects of interest (OOI), giving narrative content. To ensure that such content is not missed by the viewer, aligning the two primary zones (A and 1) is ideal. By suddenly changing the placement of the ROI, as an example from A to C, there is a lack of guidance, unless cues are utilised.



Likewise framing, editing is an essential tool utilised for traditional movies. Cuts and fades are, for example, frequently used transitions, that can extend the meaning of the narrative [1]. As for transitions in VR, they are often perceived as a change in the world experience, drawing attention to the virtuality of the experience, decreasing immersion [4]. A study [4] investigating transitions in VR, stated that teleportation and flying often broke the feeling of presence. Including transitions in a VR experience should, therefore, be kept to the minimum and should not be teleportation nor flying.

3 DESIGN

The narrative is based on cinematic techniques to show a theme, rather than having an actual story-line [1]. This was done to accommodate the atmosphere of the electronic music, which the MV was created for - Project Pablo with "Software".

The visual foundation of the created VE takes inspiration from A-Ha's "Take On Me", where the main character continuously enters different realities. The visuals were also based on cultural visual trends, showing that abstract art and vibrant colours such as proton purple, plastic pink and UFO green, were popular choices in 2019 [9, 6]. These trends were seen to comprise the art style "Vaporwave". To accommodate the art style, the following nine OOIs, whereto the viewer's attention would be guided, were chosen: arcade machines, soda bottle, TVs, mannequins, diving helmet, jukebox, plants, and a keyboard. These were not the only objects in the VE but the ones that were emphasised by the visual cues.

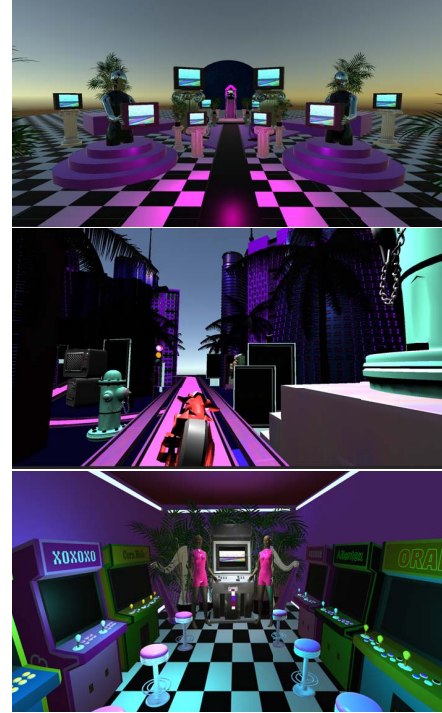


Figure 2: Some screenshots from the music video

Two design iterations were made to evaluate the OF's ability to guide the viewer's attention. From the first iteration, it was found that the cue should not necessarily have a contrasting, but noticeable colour depending on its surroundings. Secondly, the OF should have animate movement, which was more noticeable. The two OFs, a 'blob' cue and a Pac-Man cue, would follow the zone-division system. Hence moving through zones (A, then B, then C), rather than jumping zones (e.g. A to C).

4 METHODS

Initial research led to the creation of the following hypotheses:

- **H0:** A VR environment, utilising attention cues and visual trends, does not increase the viewer's level of presence for MVs.
- **H1:** A VR environment, utilising attention cues and visual trends, increases the viewer's level of presence for MVs.

Due to smartphones being the most commonly used medium to watch MVs, the VR experience will be compared to this. Both mediums will incorporate a 360° video, including either head-movement or movement of the phone as interaction.

The study was conducted within-group, involving students from Aalborg University in Copenhagen. The students were mainly from the study of Medialogy, where a common interest for VR is found. With what medium the participants would see the MV first, was randomised. Participants beginning in VR, is group VR_Ph, and participants beginning with the mobile phone is group Ph_VR, as seen in the table below:

Group name	Medium used	
Ph_VR	Phone	VR
VR_Ph	VR	Phone

After each version, the participants would answer the Slater-Usch-Steed (SUS) questionnaire to measure the level of presence in the VE [16], and a list asking what OOI were noticed. Here, all nine OOIs were listed, with the option of stating additional perceived objects. Lastly, a control question about whether the genre of music affected their responses, to see if this had an influence on the MV's ability to immerse the participant. Supporting the participants' statements, an eye tracker (Vive Pro Eye) was used to register which OOIs were actually looked at.

The researchers did not reveal the intent of the test, which left the experiment to solely rely on the eye-tracker for understanding if the OF had actually guided the viewer's attention as intended.

5 RESULTS

The MV went through two design iterations, evaluated in VR. Feedback from participants indicated that the aesthetics of the environment and the OFs were coherent with the music, giving a sense of a complete world. A final evaluation, investigating the level of presence could hence be conducted. The evaluation was conducted on music video whose design is shown in Figure 2.

5.1 Participants

The study has a total of 34 participants (24 males, 9 females, 1 other), where most were in their 20s, and one above 30 years. There were 17 subjects in each group. All apart from 3 respondents had tried VR before the study.

From the SUS questionnaire, the mean for VR was 4.69 ($median=5$, $SE=0.29$), which was significantly lower for the phone version with mean 2.08 ($median=2.25$, $SE=0.24$). The QQ-plots for the means can be seen in Figure 3, indicating a linear positive trend. The Anderson-Darling test indicated that data followed a

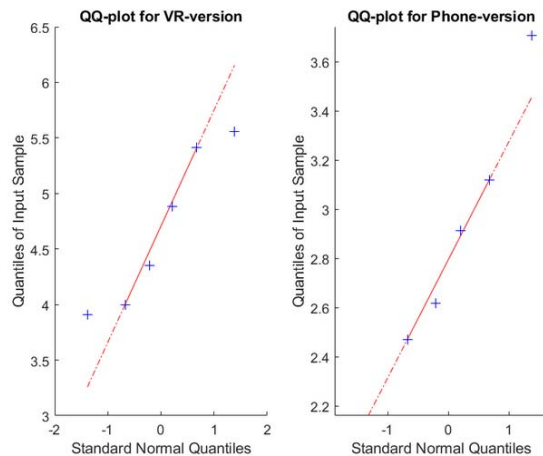


Figure 3: QQ plots

normal distribution, both VR_Ph ($p=0.50$) and Ph_VR ($p=0.97$) and Levene's test indicated furthermore homogeneous variance for both VR_Ph ($p=0.76$) and Ph_VR ($p=0.34$). Albeit a questionable internal consistency for both versions ($p=0.60$) by Cronbach's alpha, it was decided to continue with parametric testing.

The t-test indicated that there was a significant difference between the two versions ($p=3.56^{-4}$), rejecting H_0 . H_1 is therefore accepted, indicating that VR gives a higher level of presence, compared to the phone.

5.2 Eye-tracker

The eye-tracker registered that all but six OOIs were seen by all participants. These six are depicted in Figure 4. The soda bottle was registered by the least number of participants; 10 registrations were made when using VR as the first medium and 12 using VR as the second. Overall, an increase of 0-3 observed OOIs were registered when using the VR as the second medium.

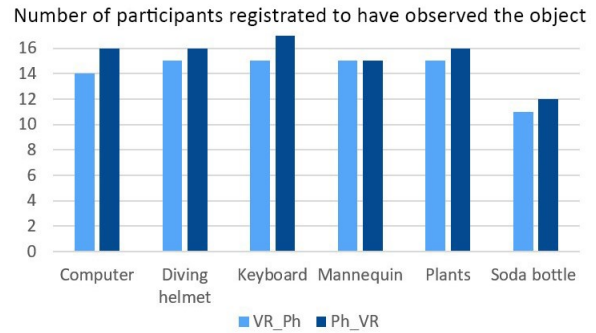


Figure 4: Number of participants for which the eye-tracker registered to have observed the OOI. The objects not displayed were seen by all participants.

5.3 Difference between Eye-tracker and Questionnaire

The subjects reported having seen all but seven OOIs. The number of reports for these seven is depicted in Figure 5. *Ph_VR phone* and *Ph_VR VR* is the Ph_VR group experiencing the MV first by phone and then VR, respectively. The biggest difference is found with the diving helmet for group Ph_VR. Here, just four reports were seen for *Ph_VR phone*, but 14 for *Ph_VR VR*.

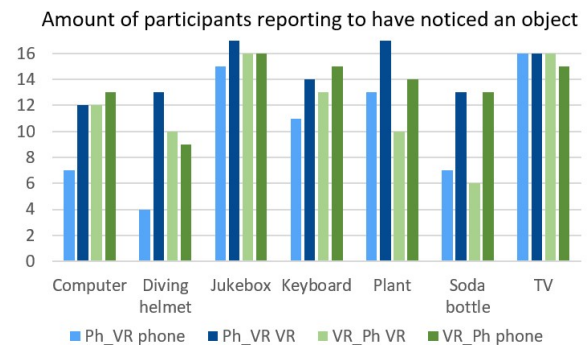


Figure 5: The number of participants reported to have perceived the OOI. Objects not graphed were reported to be seen equally amount of times in both versions, by at least 16 participants.

Between the number of reports from the subjects and the number of registrations from the eye-tracker, there is a general tendency of surplus registrations (see Figure 6). Although mannequins, plants, and soda bottle were reported more often than registered, especially for group Ph_VR.

Besides the nine listed OOIs, following additional objects were mentioned: a chicken, the blob cue, hands, harps, Pac-Man cue, pillars, radio, and a vinyl player. The Pac-Man and chicken were reported four times, compared to other being reported just once or twice. The Pac-Man was perceived for both versions, while the chicken was noticed when watching the MV the second time.

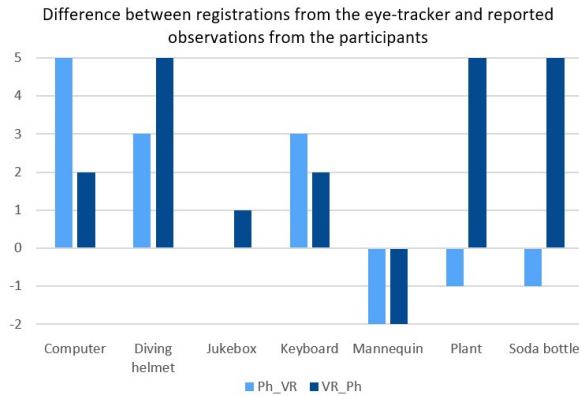


Figure 6: The difference between what the participants reported to see and what the eye-tracker measured.

6 DISCUSSION

Multiple participants reported that their sense of 'being there' was positively impacted by the OF, although it cannot be concluded that these were the key factors. Additional questions regarding the relationship between the feeling of presence and the guidance of the OF should have been included. While the OF functioned as intended for both versions, allowing narrative immersion, it was seen that further including a medium that allows for a higher level of system immersion, results in a higher sense of presence. Although, seeing the difference between the reports and registrations of perceived OOI, it is debatable whether the participants understood that these objects were important for the theme of the MV.

The procedure of having the participants answering the same questionnaire after each version may have created a bias and be the reason for the questionable reliability of the data. The participants could confuse the two experiences and hence mistakenly report to have seen certain objects just because of remembrance. Additionally, the risk of memorisation was increased since the list would reveal what objects to notice, processing information from the top-down [3]. The phrasing of some questions in the SUS questionnaire might have been misleading since it was designed for measuring presence solely within VE. Thus an alternative method for measuring presence for a 360° video watched on a mobile phone could have been used. It should be mentioned that the visual quality of the MV, was remarkably worse on phone, being decreased when uploading the 360° video to YouTube. This could further decrease the sense of presence.

7 CONCLUSION

There was significant difference in the level of presence between the VR and phone version. Results indicate that VR creates a stronger sense of presence. While the OFs were observed to function as intended, leading the viewer's attention to the OOI, it is debatable whether these were the cause of the level of presence. There was consensus on a sense of a complete world, given that the art style of the VE and incorporated objects were coherent with the atmosphere of the music. As seen, VR is thus a potentially new medium to use for artists, who want to create an immersive experience for their audience. Further investigation into how the aesthetics' coherence with the music can affect the experience of a holistic world, together with the guidance of the viewer's attention to increase the level of narrative immersion is needed.

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